

RATES OF CONVERGENCE FOR A GLOBAL MEASURE OF PERFORMANCE OF KERNEL DENSITY ESTIMATORS

Michael Falk: Department of Mathematics, University of Siegen
Hölderlinstr. 3, 5900 Siegen 21, West Germany.

Key words: Berry-Esseen theorem; conditional expectation;
U-statistics.

Summary: A result of Berry-Esseen type is proved for a global measure of performance of kernel density estimators under fairly general conditions on the underlying density. This general result is specialized to the different cases where the density is smooth on the real line as well as concentrated on a finite interval.

1. INTRODUCTION

Since their introduction by Rosenblatt (1956) and Parzen (1962), there has been a great interest in kernel estimators of the density f of a probability measure P on the real line. These estimator sequences are of the form

$$f_n(t) := (n\alpha_n)^{-1} \sum_{i=1}^n k((t-x_i)/\alpha_n)$$

where $\int k(x)dx = 1$, α_n , $n \in \mathbb{N}$, is a sequence of positive constants tending to zero as n increases and $x_1, \dots, x_n$ is a sample generated

STMA 1985 SUBJECT CLASSIFICATION: 04:180 (05:020)

independently according to P. For surveys of this particular subject we refer to the articles by Scott *et al.* (1977) and Wertz (1978).

For the investigation of the global performance of kernel density estimators several measures suggest themselves: the (weighted) supremum distance $\sup_t |f_n(t) - f(t)|/f(t)^{\frac{1}{2}}$ was considered by Bickel and Rosenblatt (1973) (see also the correction note by Bickel and Rosenblatt (1975)) and the L_1 -norm $\int |f_n(t) - f(t)| dt$ is studied in the book by Devroye and Györfi (1983), for example.

However, one particular interesting distance between probability densities f and f_0 with respective probability measures P , P_0 is given by

$$\|f - f_0\|^2 := \int (f/f_0 - 1)^2 dP_0.$$

This distance is via Taylor expansion not only closely related to the Kullback-Leibler information number $\int \log(f_0/f) dP$ but it is also asymptotically equivalent to the Hellinger distance under appropriate regularity conditions.

Moreover, $\|f - f_0\|$ proves to be an appropriate distance for product measures since it is related to the distances between their components in a simple way. For details we refer to section 6 of the monograph by Pfanzagl (1982).

Consequently, approximating the sample distribution function by an appropriate Brownian motion process Bickel and Rosenblatt (1973) proved asymptotic normality of the particular global measure of performance of kernel density estimators

$$T_n := \int_0^1 (f_n/f - 1)^2 dP = \int_0^1 (f_n(t) - f(t))^2 / f(t) dt$$

which can be regarded as a χ^2 -statistic with infinite small cells.

This result is achieved among others under the assumption that f is positive and twice differentiable on $\mathbb{R}$ with bounded second derivative

and that α_n lies between $n^{-\frac{1}{4}}$ and $n^{-\frac{2}{9}}$, roughly speaking. Notice that this condition excludes the optimal bandwidths which are approximately of order $n^{-\frac{1}{5}}$.

Moreover, as an application Bickel and Rosenblatt (1973) defined a critical region based on T_n and computed its asymptotic power on a sequence of smooth alternatives converging to the uniform distribution on $(0,1)$ which is the null-hypothesis.

However, a closer examination of Bickel and Rosenblatt's result reveals that this critical region is not of asymptotic level α but of level 1.

The deeper reason for this is that, in general, a kernel estimator is not a consistent estimator of the density at the endpoints of a distribution (see Falk (1984)). This entails that the proper standardization of T_n to achieve asymptotic normality in the case where P is concentrated on a finite interval differs completely from that which is needed in the case where the density is smooth on the real line.

Therefore, we will prove a Berry-Esseen type result for T_n under fairly general conditions on the underlying density (see Lemma 2.3 below) and we will specialize it to the different cases just mentioned.

As an application we can adjust the proper critical value in the testing problem described above and, moreover, we can indicate its accuracy. Furthermore, our approach to the investigation of T_n via U-statistics allows the application of optimal bandwidths.

2. MAIN RESULTS

Throughout this section let P be a probability measure on the real line with Lebesgue density f . We assume f to be positive on $[0,1]$, say. The following representation of T_n is obvious.

$$(2.1) \quad T_n(\underline{x})$$

$$\begin{aligned}
 &= n^{-2} \sum_{i,j=1; i \neq j}^n \int_0^1 \{ \alpha_n^{-1} k((t-x_i)/\alpha_n)/f(t) - 1 \} \\
 &\quad \cdot \{ \alpha_n^{-1} k((t-x_j)/\alpha_n)/f(t) - 1 \} P(dt) \\
 &\quad + n^{-2} \sum_{i=1}^n \int_0^1 \{ \alpha_n^{-1} k((t-x_i)/\alpha_n)/f(t) - 1 \}^2 P(dt) \\
 &=: n^{-2} \sum_{i,j=1; i \neq j}^n U_n(x_i, x_j) + n^{-2} \sum_{i=1}^n g_n(x_i) \\
 &= n^{-2} \sum_{i,j=1; i \neq j}^n \{ U_n(x_i, x_j) + 2^{-1}(n-1)^{-1} (g_n(x_i) + g_n(x_j)) \}.
 \end{aligned}$$

Thus, up to norming constants, T_n can be represented as a U -statistic. This will enable us to investigate T_n by using standard techniques for such statistics.

The conditional expectation of U_n with respect to its first co-ordinate is given by

$$h_n(x) := \int U_n(x, y) P(dy)$$

and we denote by μ_n and ν_n the expectation of h_n and g_n , respectively, assuming that these quantities exist.

Formula (2.1) implies the following representation of T_n which is essential to our considerations.

$$(2.2) \quad T_n(\underline{x})$$

$$= n^{-2} \sum_{i=1}^n \{ 2(n-1)(h_n(x_i) - \mu_n) + (g_n(x_i) - \nu_n) \} + (1-n^{-1})\mu_n + n^{-1}\nu_n$$

$$\begin{aligned}
 & + n^{-2} \sum_{i,j=1; i \neq j}^n \{U_n(x_i, x_j) - \mu_n - (h_n(x_i) - \mu_n) - (h_n(x_j) - \mu_n)\} \\
 & =: n^{-2} \sum_{i=1}^n \psi_n(x_i) + (1-n^{-1})\mu_n + n^{-1}\nu_n + R_n(\underline{x}).
 \end{aligned}$$

Hence, $T_n - (1-n^{-1})\mu_n - n^{-1}\nu_n$ is a sum of independent identically distributed random variables centered at expectation plus a remainder term which will tend to zero under appropriate conditions on f and α_n , $n \in \mathbb{N}$.

Therefore, the Berry-Esseen theorem for sums of i.i.d. random variables together with an appropriate estimate of R_n will imply our following Lemma 2.3. We remark that we make use of this particular approach since direct application to T_n of the Berry-Esseen theorem for U-statistics by Callaert and Janssen (1978) is not appropriate for our purposes since this leads to rather restrictive conditions on α_n , $n \in \mathbb{N}$.

The following result will be the main tool for proving our theorems.

Lemma 2.3

Assume that $\sigma_n^2 := E(\psi_n^2) \in (0, \infty)$. For any $r \in \mathbb{N}$ there exist universal positive constants C_1 and $C_2 = C_2(r)$ such that

$$\begin{aligned}
 & \sup_{t \in \mathbb{R}} |P^n\{n^{\frac{3}{2}}\sigma_n^{-1}(T_n - (1-n^{-1})\mu_n - n^{-1}\nu_n) \leq t\} - \Phi(t)| \\
 & \leq C_1 n^{-\frac{1}{2}} \sigma_n^{-3} E(|\psi_n|^3) + C_2 \sigma^{-2r/(2r+1)} n^{r/(2r+1)} E(U_n^{2r})^{1/(2r+1)}
 \end{aligned}$$

where Φ denotes the distribution function of the standard normal distribution $N_{(0,1)}$ and P^n the n -fold independent product of P .

Specifying the underlying density f and assuming further conditions on k and α_n , $n \in \mathbb{N}$, we shall derive in the following several corollaries to the above result. First we treat the case where f is twice continuous

differentiable on the real line.

Theorem 2.4

Assume that f is twice continuous differentiable on $\mathbb{R}$ with bounded second derivative and that $\int k(x)dx = 1$, $\int xk(x)dx = 0$, $\int x^2 |k(x)| dx < \infty$ and $\int k^2(x)dx < \infty$. Then, for any $r \in \mathbb{N}$ if $\alpha_n \xrightarrow{n \in \mathbb{N}} 0$, $n\alpha_n^3 \xrightarrow{n \in \mathbb{N}} \infty$

$$\begin{aligned} \sup_{t \in \mathbb{R}} |P^n\{n^{1/2}\alpha_n^{-2}c_n^{-1}(T_n - \mu_n - n^{-1}\nu_n) \leq t\} - \Phi(t)| \\ = O(n^{-r/(2r+1)}\alpha_n^{-(6r-1)/(2r+1)}) \end{aligned}$$

$$\text{where } c_n^2 := 4\alpha_n^{-4}E((h_n - \mu_n)^2) \xrightarrow{n \in \mathbb{N}} (\int k(x)x^2 dx)^2 \int (f''(x))^2 / f(x) dx.$$

The asymptotic values of μ_n and ν_n are given by the following two results.

Lemma 2.5

Assume that f is twice continuous differentiable on $\mathbb{R}$ with bounded second derivative and that $\int k(x)dx = 1$, $\int xk(x)dx = 0$, $\int x^2 |k(x)| dx < \infty$. Then, if $\alpha_n \xrightarrow{n \in \mathbb{N}} 0$

$$\begin{aligned} \text{(i) } \lim_{n \in \mathbb{N}} \alpha_n^{-4} E(h_n) &= \lim_{n \in \mathbb{N}} \alpha_n^{-4} E(h_n^2) \\ &= 4^{-1} (\int k(x)x^2 dx)^2 \int_0^1 (f''(x))^2 / f(x) dx =: c \end{aligned}$$

and thus,

$$\text{(ii) } \lim_{n \in \mathbb{N}} \alpha_n^{-4} E((h_n - \mu_n)^2) = c.$$

The next result will be useful also in the case where f is concentrated on $[0,1]$.

Lemma 2.6

If f is continuous on $(0,1)$, bounded away from zero and infinity on $[0,1]$, $\int k^2(x)dx < \infty$ and $\alpha_n \rightarrow 0$, then

$$(i) \lim_{n \in \mathbb{N}} \alpha_n E(g_n) = \int k^2(x)dx, \quad \lim_{n \in \mathbb{N}} \alpha_n^2 E(g_n^2) = (\int k^2(x)dx)^2 \int_0^1 f(x)^{-1} dx;$$

and thus,

$$(ii) \lim_{n \in \mathbb{N}} \alpha_n^2 E((g_n - \nu_n)^2) = (\int k^2(x)dx)^2 \left\{ \int_0^1 f(x)^{-1} dx - 1 \right\}.$$

Remark

If in addition $n\alpha_n^5 \rightarrow \infty$, then Theorem 2.4 implies that

$n^{1/2} \alpha_n^{-2} (T_n - \mu_n - n^{-1} \nu_n)$ is asymptotically normal distributed. Moreover, under the assumptions of Theorem 2.4 the optimal bandwidths according to a global measure of performance are usually of the form $d_n n^{-1/5}$ where d_n slowly tends to infinity (see, for example, section 4 in Stute (1982) or formulae (2.15), (2.16) in Reiss (1975)). Thus, such optimal sequences are included in Theorem 2.4.

Theorem 2.4 together with some additional computations entails the following law of large numbers for T_n .

Corollary 2.7

If in addition to the assumptions of Theorem 2.4 $n\alpha_n^5 \rightarrow \infty$, then

$$\alpha_n^{-4} T_n \xrightarrow{p} 4^{-1} (\int k(x)x^2 dx)^2 \int_0^1 (f''(x))^2 / f(x) dx$$

where $\xrightarrow{p}$ denotes convergence in probability.

Next we deal with the case where the density f has finite support $[0,1]$ on which it is continuous and bounded.

It is easy to see that in this situation under common assumptions on k and α_n , $n \in \mathbb{N}$, the kernel estimator usually fails at the endpoints of the support of f (see formula (2) in Falk (1984)) which implies

$$(2.8) \quad \sup_{t \in (0,1)} |f_n(t) - f(t)| \not\rightarrow 0 \text{ in probability.}$$

Thus, the statistic $\sup_{t \in (0,1)} |f_n(t) - f(t)|$ is not appropriate for

constructing a test of goodness of fit if, for example, f is the density of the uniform distribution on $(0,1)$.

However, since the set of points where f_n does not converge to f has P -measure zero, one should expect that nevertheless the statistic T_n converges to zero under P^n . This suggests the construction of a test of goodness of fit based on T_n . These ideas are verified in our next results.

Theorem 2.9

Let f have support $[0,1]$ on which it is continuous and positive. Assume further that f is continuous differentiable on $(0,1)$ with bounded derivative and that k is bounded and satisfies $\int k(x) dx = 1$, $\int x k(x) dx = 0$, $\int |x|^{1+\delta} k(x) dx < \infty$ for some $\delta > 0$ and $\int k^2(x) dx < \infty$. Then, for any $r \in \mathbb{N}$ if $\alpha_n \rightarrow 0$ and $n\alpha_n^{3/2} \rightarrow \infty$

$$\begin{aligned} \sup_{t \in \mathbb{N}} |P_n^{1/2} \alpha_n^{-1/2} c_n^{-1} (T_n - \mu_n - n^{-1} \nu_n) \leq t) - \Phi(t)| \\ = O(n^{-r/(2r+1)} \alpha_n^{-(3r-1)/(2r+1)}), \end{aligned}$$

$$\text{where } c_n^2 := 4\alpha_n^{-1} E((h_n - \mu_n)^2) \xrightarrow[n \in \mathbb{N}]{} 4f(0) \int_0^\infty \left\{ \int_{-x}^\infty k(t)(1-K(t+x)) dt \right\}^2 dx$$

$$+ 4f(1) \int_{-\infty}^0 \left\{ \int_{-\infty}^{-x} k(t)K(t+x)dt \right\}^2 dx < \infty, \quad K(x) := \int_{-\infty}^x k(y)dy.$$

The asymptotic value of ν_n is given in Lemma 2.6 whereas the asymptotic value of μ_n under the specific assumptions on f stated in Theorem 2.9 is computed in the following result.

Lemma 2.10

Let f have support $[0,1]$ on which it is positive. Assume that f is continuous differentiable on $(0,1)$ with bounded derivative and let k satisfy $\int k(x)dx = 1$, $\int xk(x)dx = 0$ and $\int |x|^{1+\delta} k(x)|dx < \infty$ for some $\delta > 0$. Then, if $\alpha_n \rightarrow 0$
 $n \in \mathbb{N}$

$$(i) \quad \lim_{n \in \mathbb{N}} \alpha_n^{-1} E(h_n) = f(0) \int_0^\infty (K(t)-1)^2 dt + f(1) \int_{-\infty}^0 K(t)^2 dt < \infty$$

and

$$(ii) \quad \lim_{n \in \mathbb{N}} \alpha_n^{-1} E(h_n^2) = f(0) \int_0^\infty \left\{ \int_{-x}^\infty k(t)(1-K(t+x))dt \right\}^2 dx \\ + f(1) \int_{-\infty}^0 \left\{ \int_{-\infty}^{-x} k(t)K(t+x)dt \right\}^2 dx < \infty.$$

As a consequence we get the following law of large numbers for T_n in the case where f is concentrated on $[0,1]$.

Corollary 2.11

If in addition to the assumptions of Theorem 2.9 $\alpha_n^2 \rightarrow 0$, then
 $n \in \mathbb{N}$

$$\alpha_n^{-1} T_n \xrightarrow{p^n} f(0) \int_0^\infty (K(t)-1)^2 dt + f(1) \int_{-\infty}^0 K(t)^2 dt < \infty.$$

3. APPLICATIONS

Take for example $k(x) = 1_{[-1/2, 1/2]}(x)$ and P the uniform distribution on $(0,1)$, denoted by Q ; then

$$T_n = \int_0^1 \left\{ (n\alpha_n)^{-1} N[t - \alpha_n/2, t + \alpha_n/2] - 1 \right\}^2 dt$$

where $N_{[x,y]}$ denotes the number of observations falling into the interval $[x,y]$.

It was shown by Bickel and Rosenblatt (1973), page 1083, that $V_n := \alpha_n^{-1/2}((n\alpha_n)T_n - 1)$ is asymptotically normal distributed under a sequence of smooth alternatives $P_n, n \in \mathbb{N}$, converging to Q .

These alternatives have Lebesgue densities $g_n(x) = 1_{[0,1]}(x) + \gamma_n \eta(x) + o(\gamma_n)$, $n \in \mathbb{N}$, where the sequence $g_n, n \in \mathbb{N}$, satisfies some smoothness conditions and $\gamma_n = n^{-1/2+\delta/4}$, $\delta < 1/4$, $\alpha_n = n^{-\delta}$.

Furthermore, Bickel and Rosenblatt computed the asymptotic power function of a critical region C_n based on V_n on P_n as well as its asymptotic relative efficiency with respect to a χ^2 test of goodness of fit which is rather similar to that based on V_n .

But their standardization of T_n is not adapted to the hypothesis Q which entails that C_n is not of asymptotic level $\alpha \in (0,1)$ but of level 1 which can be seen as follows.

Our Theorem 2.9 states that $\tilde{V}_n := n^{1/2}\alpha_n^{-1/2}(T_n - \mu_n - n^{-1}\nu_n)$ is asymptotically normal under Q^n if $n\alpha_n^2 \xrightarrow{n \in \mathbb{N}} \infty$. This result together with Lemmas 2.4 and 2.8 imply that

$$V_n = \tilde{V}_n + n^{1/2}\alpha_n^{-1/2}(n^{1/2}\alpha_n - 1)T_n + \alpha_n^{-1/2}(n^{1/2}\mu_n - 1) + (n\alpha_n)^{-1/2}\nu_n$$

tends in probability to infinity under Q^n if $n\alpha_n^2 \xrightarrow{n \in \mathbb{N}} \infty$. This yields the assertion.

On the other hand, if we interchange the null-hypothesis and the alternative, i.e. if we test $H_{0n} := \{P_n\}$ against $H_1 := \{Q\}$, then our considerations imply that the suitable critical region of asymptotic level $\alpha \in (0,1)$ on H_{0n} , $n \in \mathbb{N}$, based on V_n has asymptotic power 1 on H_1 .

4. AUXILIARY RESULTS AND PROOFS

We start with the proof of Lemma 2.3. Representation (2.2) yields for $\varepsilon > 0$

$$\begin{aligned} & \sup_{t \in \mathbb{R}} |P^n\{n^{3/2}\sigma_n^{-1}(T_n - (1-n^{-1})\mu_n - n^{-1}\nu_n) \leq t\} - \Phi(t)| \\ & \leq \sup_{t \in \mathbb{R}} |P^n\{n^{-1/2}\sigma_n^{-1} \sum_{i=1}^n \psi_n(x_i) \leq t\} - \Phi(t)| \\ & + \sup_{t \in \mathbb{R}} |\Phi(t+\varepsilon) - \Phi(t)| + P^n\{|R_n| > \varepsilon n^{-3/2}\sigma_n\}. \end{aligned}$$

Now the proof of Theorem 2.1 in Grams and Serfling (1973) shows for any $r \in \mathbb{N}$ the existence of a universal constant $D_r > 0$ such that $P^n\{|R_n| > \varepsilon n^{-3/2}\sigma_n\} \leq D_r \varepsilon^{-2r} n^{2r} \sigma_n^{-2r} E(U_n^{2r})$. Thus, by choosing

$\varepsilon := \{D_r n^{2r} \sigma_n^{-2r} E(U_n^{2r})\}^{1/(2r+1)}$ the assertion is immediate from the classical Berry-Esseen theorem for i.i.d. random variables.

4.1 Proof of Lemma 2.5

Fubini's theorem, the substitution $y = t - \alpha_n x$ and Taylor expansion yield that $E(h_n) = \int_0^1 f(t)^{-1} \{(\alpha_n^2/2) \cdot \int k(x)x^2 f''(t - \theta \alpha_n x) dx\}^2 dt$. The dominated convergence theorem now implies the first assertion. Furthermore,

$$E(h_n^2)$$

$$\begin{aligned}
&= \int_0^1 \left\{ \int_{-x/\alpha_n}^{(1-x)/\alpha_n} [k(t)f(\alpha_n t+x)^{-1} - \alpha_n] [(\alpha_n^2/2) \int k(y)y^2 f''(x+\alpha_n(t-\theta y)) dy] dt \right\}^2 \\
&\quad \cdot f(x) dx \\
&+ \int_{[0,1]^c} \left\{ \int_{-x/\alpha_n}^{(1-x)/\alpha_n} [k(t)f(\alpha_n t+x)^{-1} - \alpha_n] [(\alpha_n^2/2) \int k(y)y^2 f''(x+\alpha_n(t-\theta y)) dy] \right. \\
&\quad \left. dt \right\}^2 f(x) dx
\end{aligned}$$

$=: I_n + II_n$. The dominated convergence theorem now implies

$$\lim_{n \in \mathbb{N}} \alpha_n^{-4} I_n = c \quad \text{and} \quad \lim_{n \in \mathbb{N}} \alpha_n^{-4} II_n = 0 \quad \text{which completes the proof.}$$

In an analogous way one proves Lemma 2.6 and the next result.

Lemma 4.2

Under the assumptions of Lemma 2.5 we have $E(|h_n|^3) = O(\alpha_n^6)$ and under the assumptions of Lemma 2.6 $E(|g_n|^3) = O(\alpha_n^3)$.

We are now ready to prove Theorem 2.4.

4.3 Proof of Theorem 2.4

We want to apply Lemma 2.3. To this end we need the following representation of α_n^2 which is immediate from Lemmas 2.5 and 2.6:

$$(4.4) \quad \sigma_n^2 / (4n^2 \alpha_n^4 c_n) = 1 + O(n^{-1} \alpha_n^{-3}).$$

Furthermore, we derive from Lemmas 2.5, 2.6 and 4.2 and the fact that $\alpha_n^3 \xrightarrow{n \in \mathbb{N}} \infty$

$$(4.5) \quad E(|\psi_n^3|) = O(n^3 \alpha_n^6).$$

It is easy to check that for any $r \in \mathbb{N}$, $E(U_n^{2r}) = O(\alpha_n^{-2r+1})$. Thus, application of Lemma 2.3 together with (4.4) and (4.5) yields

$$\begin{aligned} & \sup_{t \in \mathbb{R}} P^n \{ n^{3/2} \sigma_n^{-1} (T_n - (1-n^{-1})\mu_n - n^{-1}\nu_n) \leq t \} - \Phi(t) | \\ &= O(n^{-1/2}) + O(n^{-r/(2r+1)} \alpha_n^{-(6r-1)/(2r+1)}). \end{aligned}$$

Since $n^{3/2} \sigma_n^{-1} n^{-1} \mu_n = O(n^{-1/2} \alpha_n^2)$ by (4.4) and Lemma 2.5, the assertion of Theorem 2.2 is immediate.

4.6 Proof of Corollary 2.7

We have $\alpha_n^{-4} T_n = n^{-1/2} \alpha_n^{-2} \{ n^{1/2} \alpha_n^{-2} (T_n - \mu_n - n^{-1} \nu_n) \} + \alpha_n^{-4} \mu_n + \alpha_n^{-4} n^{-1} \nu_n$.

Thus, the assertion is immediate from Theorem 2.4 and Lemmas 2.5 and 2.6.

4.7 Proof of Lemma 2.10

Fubini's theorem, the substitution $y = t - \alpha_n x$ and Taylor expansion yield

$$\begin{aligned} E(h_n) &= \int_0^1 f(t)^{-1} \left\{ \alpha_n \int_{(t-1)/\alpha_n}^{t/\alpha_n} x k(x) f'(t - \theta \alpha_n x) dx - f(t) [1 - K(t/\alpha_n) \right. \\ &\quad \left. + K((t-1)/\alpha_n)] \right\}^2 dt. \end{aligned}$$

Thus, the assertion follows from the dominated convergence theorem if we show

$$\begin{aligned} & \alpha_n^{-1} \int_0^1 f(t) \left\{ 1 - K(t/\alpha_n) + K((t-1)/\alpha_n) \right\}^2 dt \\ & \xrightarrow{n \in \mathbb{N}} f(0) \int_0^\infty (1 - K(t))^2 dt + f(1) \int_{-\infty}^0 K(t)^2 dt. \end{aligned}$$

Notice that the assumption $\int |xk(x)| dx < \infty$ implies $(1-K(t))^2 = O(t^{-2})$ as $t \rightarrow \infty$ and, equally, $K(t)^2 = O(t^{-2})$ as $t \rightarrow -\infty$. Thus, the limit value in the above result is finite. Since

$$\begin{aligned} & \alpha_n^{-1} \int_0^1 f(t) \left\{ 1 - K(t/\alpha_n) + K((t-1)/\alpha_n) \right\}^2 dt \\ &= \int_0^{1/\alpha_n} f(\alpha_n t) (1-K(t))^2 dt + \int_{-1/\alpha_n}^0 f(1+\alpha_n t) K(t)^2 dt \\ &+ 2 \int_0^{1/\alpha_n} f(\alpha_n t) (1-K(t)) K(t-\alpha_n^{-1}) dt, \end{aligned}$$

the assertion is immediate from the dominated convergence theorem.

Next we compute $E(h_n^2)$.

$$\begin{aligned} E(h_n^2) &= \int_0^1 \left\{ \int_0^1 \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) \alpha_n^{-1} k((t-y)/\alpha_n) f(y)/f(t) dt dy \right. \\ &- \left. \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) dt - \int_0^1 \alpha_n^{-1} k((t-y)/\alpha_n) f(y) dt dy + 1 \right\}^2 P(dx) \\ &=: \int_0^1 \{ a_n(x) - b_n(x) - c_n(x) + 1 \}^2 P(dx). \end{aligned}$$

By the same arguments as employed in the proof of Lemma 2.5 we derive

$$a_n(x) = \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) \{ K(t/\alpha_n) - K((t-1)/\alpha_n) \} dt + O(\alpha_n) \quad \text{and}$$

$$b_n(x) = K((1-x)/\alpha_n) - K(-x/\alpha_n). \quad \text{Furthermore,}$$

$$c_n(x) = \int_0^1 f(t) \{ K(t/\alpha_n) - K((t-1)/\alpha_n) \} dt + O(\alpha_n).$$

Denote by F the distribution function of P . Then integration by

parts yields

$$\begin{aligned}
 & \int_0^1 f(t) \left\{ K(t/\alpha_n) - K((t-1)/\alpha_n) \right\} dt \\
 &= K(\alpha_n^{-1}) - K(0) - \int_0^{\alpha_n^{-1}} F(\alpha_n t) k(t) dt + \int_{-\alpha_n^{-1}}^0 F(1+\alpha_n t) k(t) dt \\
 &= 1 + O(\alpha_n) + \int_{-\alpha_n^{-1}}^0 k(t) dt - K(0) = 1 + O(\alpha_n).
 \end{aligned}$$

Thus, $c_n = 1 + O(\alpha_n)$ and hence we have

$$\begin{aligned}
 & E(h_n^2) \\
 &= \int_0^1 \left\{ \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) [K(t/\alpha_n) - 1 - K((t-1)/\alpha_n)] dt + O(\alpha_n) \right\}^2 P(dx).
 \end{aligned}$$

We shall prove

$$\begin{aligned}
 (4.8) \quad & \alpha_n^{-1} \int_0^1 \left\{ \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) [K(t/\alpha_n) - 1 - K((t-1)/\alpha_n)] dt \right\}^2 P(dx) \\
 & \xrightarrow{n \in \mathbb{N}} f(0) \int_0^\infty \left\{ \int_{-x}^\infty k(t) (K(t+x) - 1) dt \right\}^2 dx + f(1) \int_{-\infty}^0 \left\{ \int_{-x}^\infty k(t) K(t+x) dt \right\}^2 dx < \infty
 \end{aligned}$$

which implies the assertion.

Ad (4.8)

Repeated substitution leads to

$$\alpha_n^{-1} \int_0^1 \left\{ \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) [K(t/\alpha_n) - 1] dt \right\}^2 f(x) dx$$

$$\begin{aligned}
&= \int_0^{1/\alpha_n} \left\{ \int_{-x}^{\alpha_n^{-1}-x} k(t)[K(t+x)-1]dt \right\}^2 f(\alpha_n x) dx \\
&\xrightarrow{n \in \mathbb{N}} f(0) \int_0^\infty \left\{ \int_{-x}^\infty k(t)[K(t+x)-1]dt \right\}^2 dx,
\end{aligned}$$

by the dominated convergence theorem if we show

$$(4.9) \quad \int_0^\infty \left\{ \int_{-x}^\infty |k(t)[K(t+x)-1]| dt \right\}^2 dx < \infty.$$

$$\begin{aligned}
\text{Now, this term is bounded by } &\int_0^\infty x^{-2} \left\{ |x| \int_{-x}^{-x/2} |k(t)[K(t+x)-1]| dt + \right. \\
&|x| \int_{x/2}^\infty |k(t-x)[K(t)-1]| dt \left. \right\}^2 dx \leq \int_0^\infty x^{-2} \left\{ C \int_{-x}^{-x/2} |tk(t)| dt + \right. \\
&2 \int_{x/2}^\infty |k(t-x)| \left| \int_t^\infty |yk(y)| dy \right| dt \left. \right\}^2 dx < \infty, \quad \text{where } C := 2 \int |k(x)| dx.
\end{aligned}$$

A similar argumentation yields

$$\begin{aligned}
&\alpha_n^{-1} \int_0^1 \left\{ \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) K((t-1)/\alpha_n) dt \right\}^2 P(dx) \\
&\xrightarrow{n \in \mathbb{N}} f(1) \int_{-\infty}^0 \left\{ \int_{-\infty}^{-x} k(t) K(t+x) dt \right\}^2 dx < \infty.
\end{aligned}$$

Thus, to prove (4.8) it remains to show

$$\begin{aligned}
&\alpha_n^{-1} \int_0^1 \left\{ \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) [K(t/\alpha_n)-1] dt \right\} \\
&\cdot \left\{ \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n) K((t-1)/\alpha_n) dt \right\} P(dx) \xrightarrow{n \in \mathbb{N}} 0.
\end{aligned}$$

Repeated substitution yields that this term is bounded by

$$\begin{aligned} & \int_0^{\alpha_n^{-1}} \left\{ \int_{-x}^{\alpha_n^{-1}-x} |k(t)[K(t+x)-1]| dt \right\} \left\{ \int_{-x}^{\alpha_n^{-1}-x} |k(t)K(t+x-\alpha_n^{-1})| dt \right\} f(\alpha_n x) dx \\ & \leq \int_0^{\alpha_n^{-1}} \left\{ \int_{-x}^{\infty} |k(t)[K(t+x)-1]| dt \right\} \left\{ \int_{-x}^{\alpha_n^{-1}-x} |k(t)K(t+x-\alpha_n^{-1})| dt \right\} f(\alpha_n x) dx \end{aligned}$$

→ 0 by the dominated convergence theorem since
 $n \in \mathbb{N}$

$$\int_{-\infty}^{\alpha_n^{-1}-x} |k(t)K(t+x-\alpha_n^{-1})| dt \rightarrow 0 \quad \text{for any } x > 0 \quad \text{and } n \in \mathbb{N}$$

$$\int_0^{\infty} \int_{-x}^{\infty} |k(t)[K(t+x)-1]| dt dx < \infty \quad \text{by the assumption } \int |x|^{1+\delta} k(x) dx < \infty$$

for some $\delta > 0$ as is easy to check.

Lemma 4.10

$E(|h_n|^3) = O(\alpha_n)$ under the assumptions of Lemma 2.10.

Proof

The proof of Lemma 2.10 implies $E(|h_n|^3) \leq \int_0^1 \int_0^1 \alpha_n^{-1} k((t-x)/\alpha_n)$

$\{K(t/\alpha_n) - 1 - K((t-1)/\alpha_n)\} dt\}^3 P(dx) + O(\alpha_n^2)$. The assertion then follows by the same arguments as employed in the proof of Lemma 2.10.

Proof of Theorem 2.9

The argumentation is completely analogous to that in the proof of Theorem 2.4. According to Lemmas 2.6 and 2.10

$$(4.11) \quad \sigma_n^2 / (4n^2 \alpha_n c_n) = 1 + o(n^{-1} \alpha_n^{-3/2}).$$

Furthermore, in this situation since $n \alpha_n^{4/3} \rightarrow \infty$ as $n \in \mathbb{N}$

$$(4.12) \quad E(|\psi_n|^3) = o(n^3 \alpha_n).$$

Application of Lemma 2.3 together with (4.11), (4.12) and the fact that for any $r \in \mathbb{N}$ $E(U_n^{2r}) = o(\alpha_n^{-2r+1})$ yields

$$\begin{aligned} & \sup_{t \in \mathbb{R}} |P^n \{ n^{3/2} \sigma_n^{-1} (T_n - (1 - n^{-1}) \mu_n - n^{-1} \nu_n) \leq t \} - \Phi(t)| \\ &= o(n^{-1/2} \alpha_n^{-1/2}) + o(n^{-r/(2r+1)} \alpha_n^{-(3r-1)/(2r+1)}) \\ &= o(n^{-r/(2r+1)} \alpha_n^{-(3r-1)/(2r+1)}). \end{aligned}$$

Since $n^{3/2} \sigma_n^{-1} \mu_n = o(n^{-1/2} \alpha_n^{1/2})$ by (4.11) and Lemma 2.10, the assertion of Theorem 2.9 is immediate.

ACKNOWLEDGMENT

The author would like to thank R.-D. Reiss for stimulating this research work.

REFERENCES

- BICKEL, P.J. and ROSENBLATT, M. (1973). On some global measures of the deviations of density function estimates. *Ann. Statist.* **1**, 1071-1095.
- BICKEL, P.J. and ROSENBLATT, M. (1975). Corrections to "On some global measures of the deviations of density function estimates". *Ann. Statist.* **3**, 1370.
- CALLAERT, H. and JANSSEN, P. (1978). The Berry-Esseen theorem for U-statistics. *Ann. Statist.* **6**, 153-160.
- DEVROYE, L. and GYÖRFI, L. (1983). *Non-parametric density estimation: The L_1 -view*. Wiley, New York.
- FALK, M. (1984). Kernel estimation of a density in an unknown endpoint of its support, *South African Statist. J.* **18**, 91-96.
- GRAMS, W.F. and SERFLING, R.J. (1973). Convergence rates for U-statistics and related statistics. *Ann. Statist.* **1**, 153-160.
- PARZEN, E. (1962). On estimation of a probability density function and mode. *Ann. Math. Statist.* **33**, 1065-1076.
- PFANZAGL, J. (1982). *Contributions to a general asymptotic statistical theory*. Springer, Berlin.
- REISS, R.-D. (1975). Consistency of a certain class of empirical density functions. *Metrika* **22**, 189-203.
- ROSENBLATT, M. (1956). Remarks on some non-parametric estimates of a density function. *Ann. Math. Statist.* **27**, 832-837.
- SCOTT, D.W., TAPIA, R.A. and THOMPSON, J.R. (1977). Kernel density estimation revisited. *Nonlinear Anal.* **1**, 339-372.
- STUTE, W. (1982). The oscillation behaviour of empirical processes. *Ann. Prob.* **10**, 86-107.
- WERTZ, W. (1978). *Statistical density estimation. A survey*. Vandenhoeck and Ruprecht, Göttingen.